Power

AFE 1-2

Pulsers 1-4

Transducer interconnect

AFE0_DATA0p
AFE0_DATA0n
AFE0_DATA1p
AFE0_DATA1n
AFE0_DATA2p
AFE0_DATA2n
AFE0_DATA3p
AFE0_DATA3n
AFE0_DATA4p
AFE0_DATA4n
AFE0_DATA5p
AFE0_DATA5n
AFE0_DATA6p
AFE0_DATA6n
AFE0_DATA7p
AFE0_DATA7n
AFE1_DATA0p
AFE1_DATA0n
AFE1_DATA1p
AFE1_DATA1n
AFE1_DATA2p
AFE1_DATA2n
AFE1_DATA3p
AFE1_DATA3n
AFE1_DATA4p
AFE1_DATA4n
AFE1_DATA5p
AFE1_DATA5n
AFE1_DATA6p
AFE1_DATA6n
AFE1_DATA7p
AFE1_DATA7n
AFE_CLKp
AFE_CLKn
AFE0_FCOp
AFE0_FCOn
AFE0_DCOp
AFE0_DCOn
AFE1_FCOp
AFE1_FCOn
AFE1_DCOp
AFE1_DCOn
DAC0_CSn_3V3
DAC1_CSn_3V3
DAC_D0_3V3
DAC_D1_3V3
DAC_D2_3V3
DAC_D3_3V3
DAC_D4_3V3
DAC_D5_3V3
DAC_D6_3V3
DAC_D7_3V3

AX0 AX1 AX2 AX3 AX4 AX5 AX6 AX7 AX8 AX9 AX10 AX11 AX12 AX13 AX14 AX15

TX0 TX1 TX2 TX3 TX4 TX5 TX6 TX7 TX8 TX9 TX10 TX11 TX12 TX13 TX14 TX15

CH0 CH1 CH2 CH3 CH4 CH5 CH6 CH7 CH8 CH9 CH10 CH11 CH12 CH13 CH14 CH15

PLSR0_CLK_3V3
PLSR1_CLK_3V3
PLSR2_CLK_3V3
PLSR3_CLK_3V3

PLSR0_POS0_3V3
PLSR0_POS1_3V3
PLSR0_POS2_3V3
PLSR0_POS3_3V3

PLSR0_NEG0_3V3
PLSR0_NEG1_3V3
PLSR0_NEG2_3V3
PLSR0_NEG3_3V3

PLSR0_SEL0_3V3
PLSR0_SEL1_3V3
PLSR0_SEL2_3V3
PLSR0_SEL3_3V3

PLSR1_POS0_3V3
PLSR1_POS1_3V3
PLSR1_POS2_3V3
PLSR1_POS3_3V3

PLSR1_NEG0_3V3
PLSR1_NEG1_3V3
PLSR1_NEG2_3V3
PLSR1_NEG3_3V3

PLSR1_SEL0_3V3
PLSR1_SEL1_3V3
PLSR1_SEL2_3V3
PLSR1_SEL3_3V3

PLSR2_POS0_3V3
PLSR2_POS1_3V3
PLSR2_POS2_3V3
PLSR2_POS3_3V3

PLSR2_NEG0_3V3
PLSR2_NEG1_3V3
PLSR2_NEG2_3V3
PLSR2_NEG3_3V3

PLSR2_SEL0_3V3
PLSR2_SEL1_3V3
PLSR2_SEL2_3V3
PLSR2_SEL3_3V3

PLSR3_POS0_3V3
PLSR3_POS1_3V3
PLSR3_POS2_3V3
PLSR3_POS3_3V3

PLSR3_NEG0_3V3
PLSR3_NEG1_3V3
PLSR3_NEG2_3V3
PLSR3_NEG3_3V3

PLSR3_SEL0_3V3
PLSR3_SEL1_3V3
PLSR3_SEL2_3V3
PLSR3_SEL3_3V3

J4A J4B J4C J4D J4E J4F — DL1-156RW-6B

J23 — 0022112032 — 12V — FAN_FREQ — TP63

J21 — 15469274 — 80V 40V 40V

J22 — 15469274 — -40V -40V -80V

Label each post with Voltage

J1 — TSM-112-02-L-DV-LC

PLSR_REN
PLSR_OEN
PLSR0_OTPn
PLSR1_OTPn
PLSR2_OTPn
PLSR3_OTPn
AFE_SDIO
AFE_SCK
AFE0_CSn
AFE1_CSn

#4 MTG1 #4 MTG2 #4 MTG3 #4 MTG4 #4 MTG5

AFE_TRIGp AFE_TRIGn AFE_SDIO AFE_SCLK AFE0_nCS AFE1_nCS

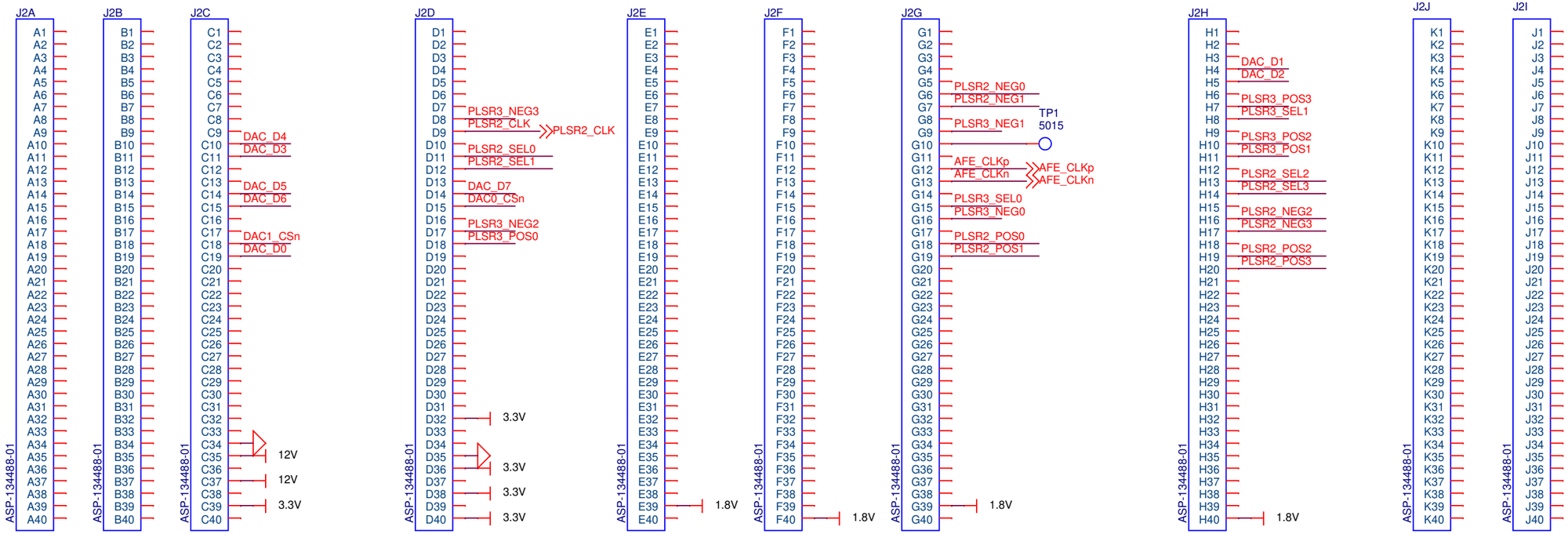
PLSR_REN_3V3
PLSR_OEN_3V3

J4G — MT1 MT2 MT3 MT4 — DL1-156RW-6B

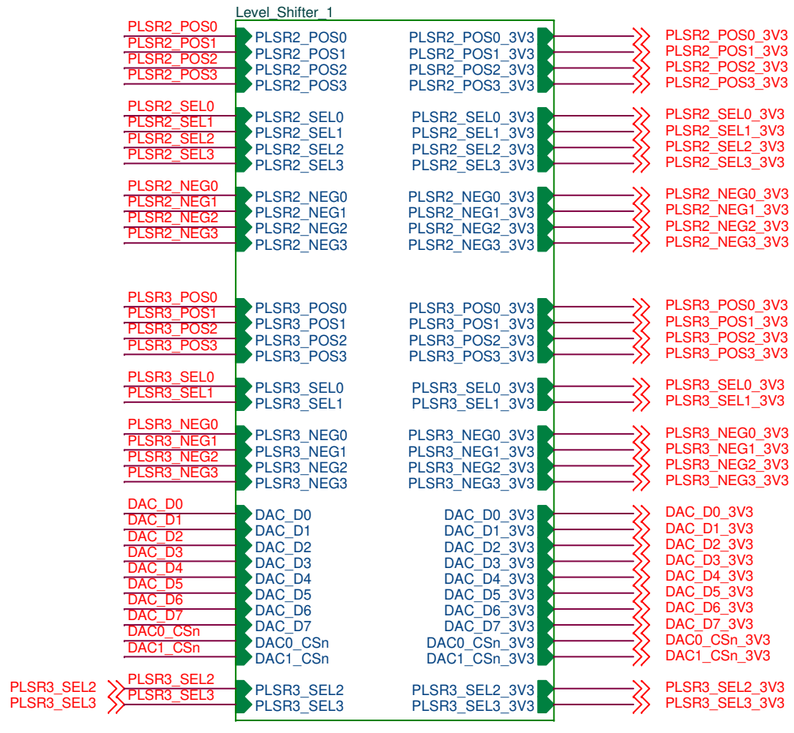
Released Rpaddock 3/12/19

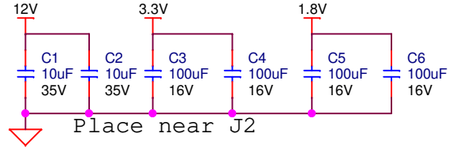
Creare
16 Great Hollow Road
Hanover, NH 03755

TITLE: Open Image - Interloper - Top A			
SIZE: B	CAGE NO.: 8A287	DOC. NO.: 1010260-900-0011	REV: A
DRAWN BY: RPaddock		ECO: <ECO>	
DATE: Thursday, March 12, 2020		SHEET 1 OF 11	



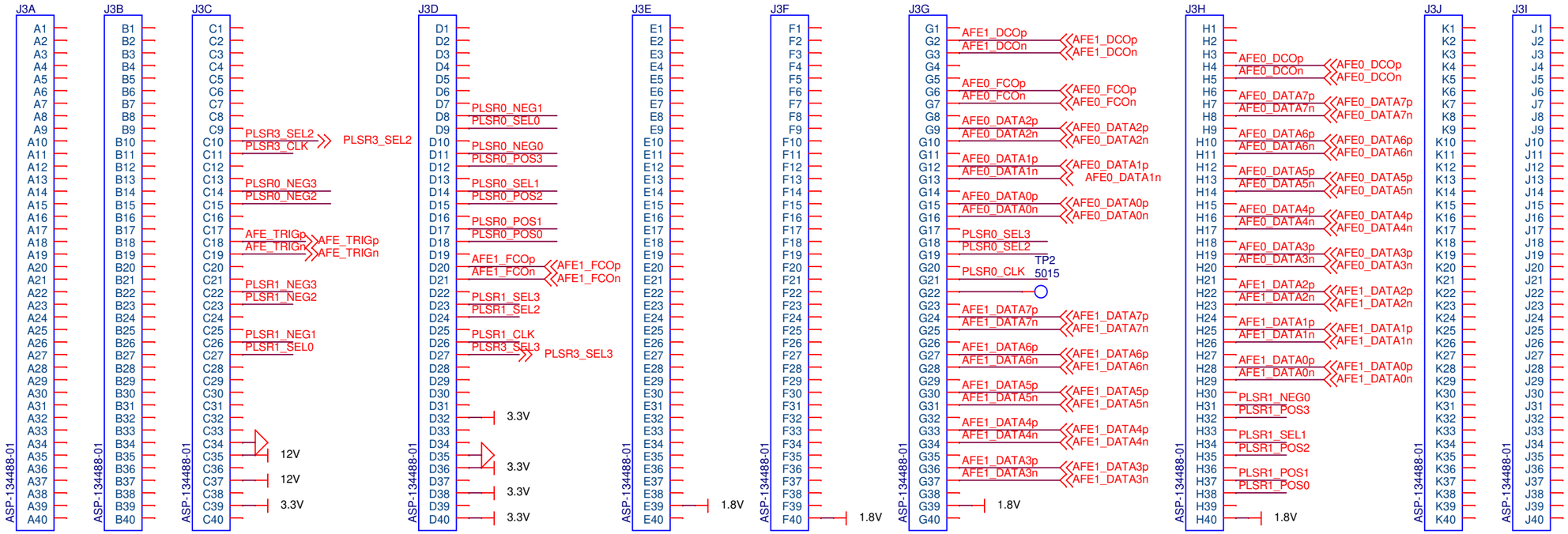
ZCU106 J4 Mate ^



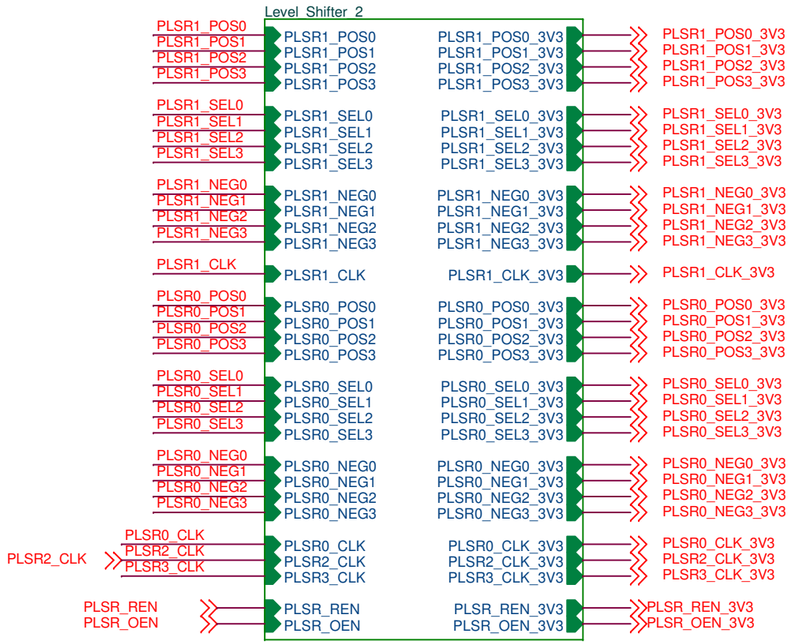


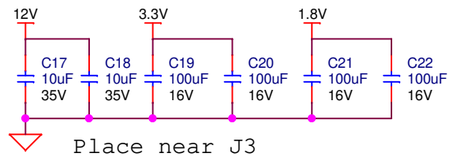
Creare
16 Great Hollow Road
Hanover, NH 03755

TITLE: Open Image - Interloper - Top B			
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DRAWN BY: RPaddock		ECO: <ECO>	
DATE: Thursday, March 12, 2020		SHEET 2 OF 11	



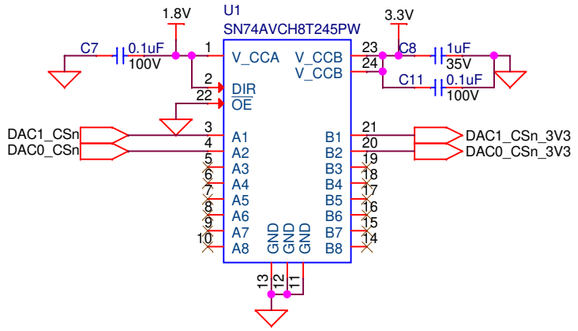
ZCU106 J5 Mate^

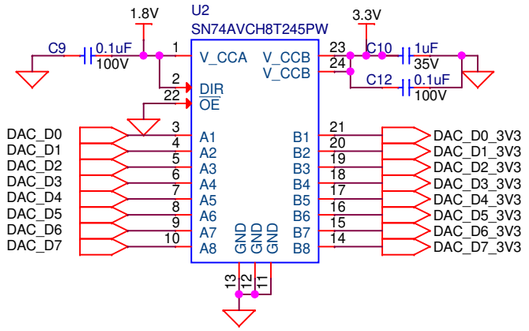


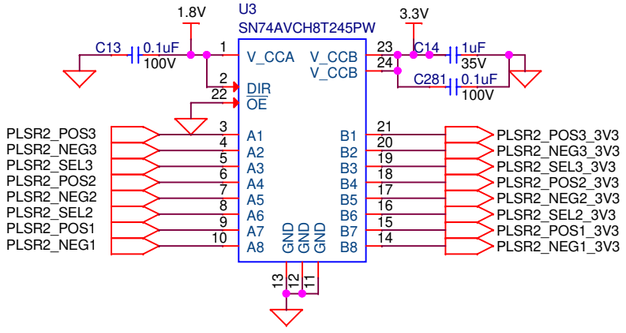


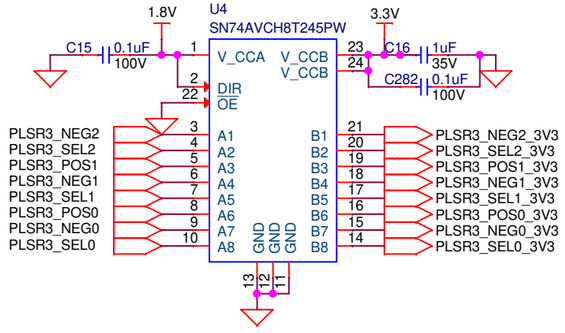
Creare
16 Great Hollow Road
Hanover, NH 03755

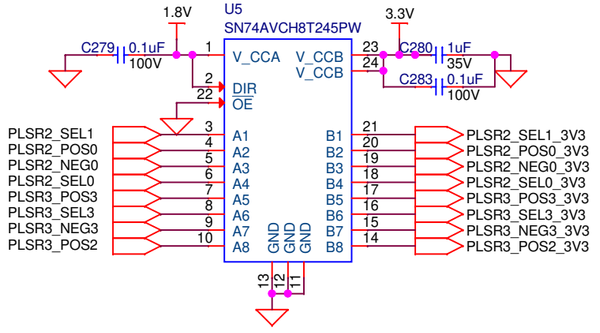
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DATE: Thursday, March 12, 2020		SHEET 3 OF 11	





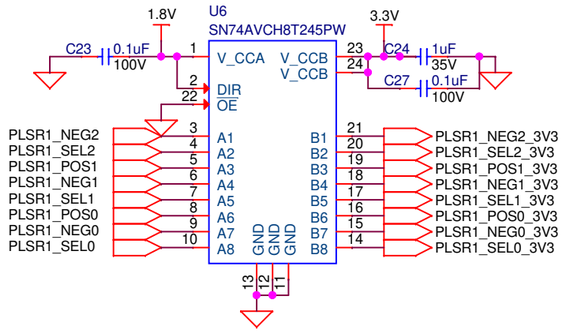


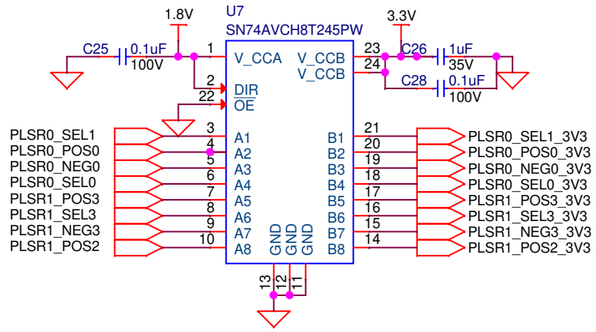


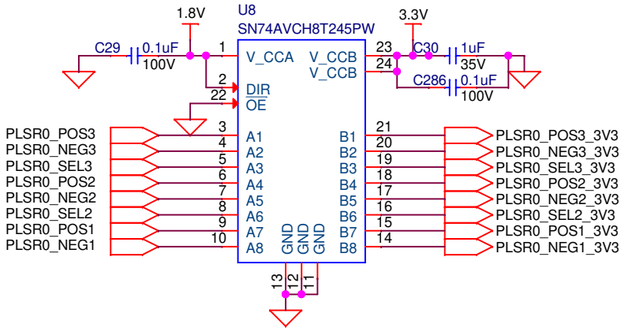


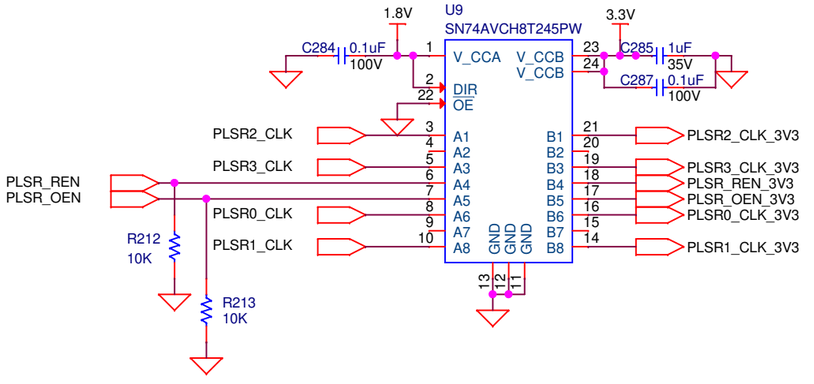
Creare
16 Great Hollow Road
Hanover, NH 03755

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DRAWN BY: RPaddock		ECO: <ECO>	
DATE: Tuesday, March 03, 2020		SHEET 4 OF 11	



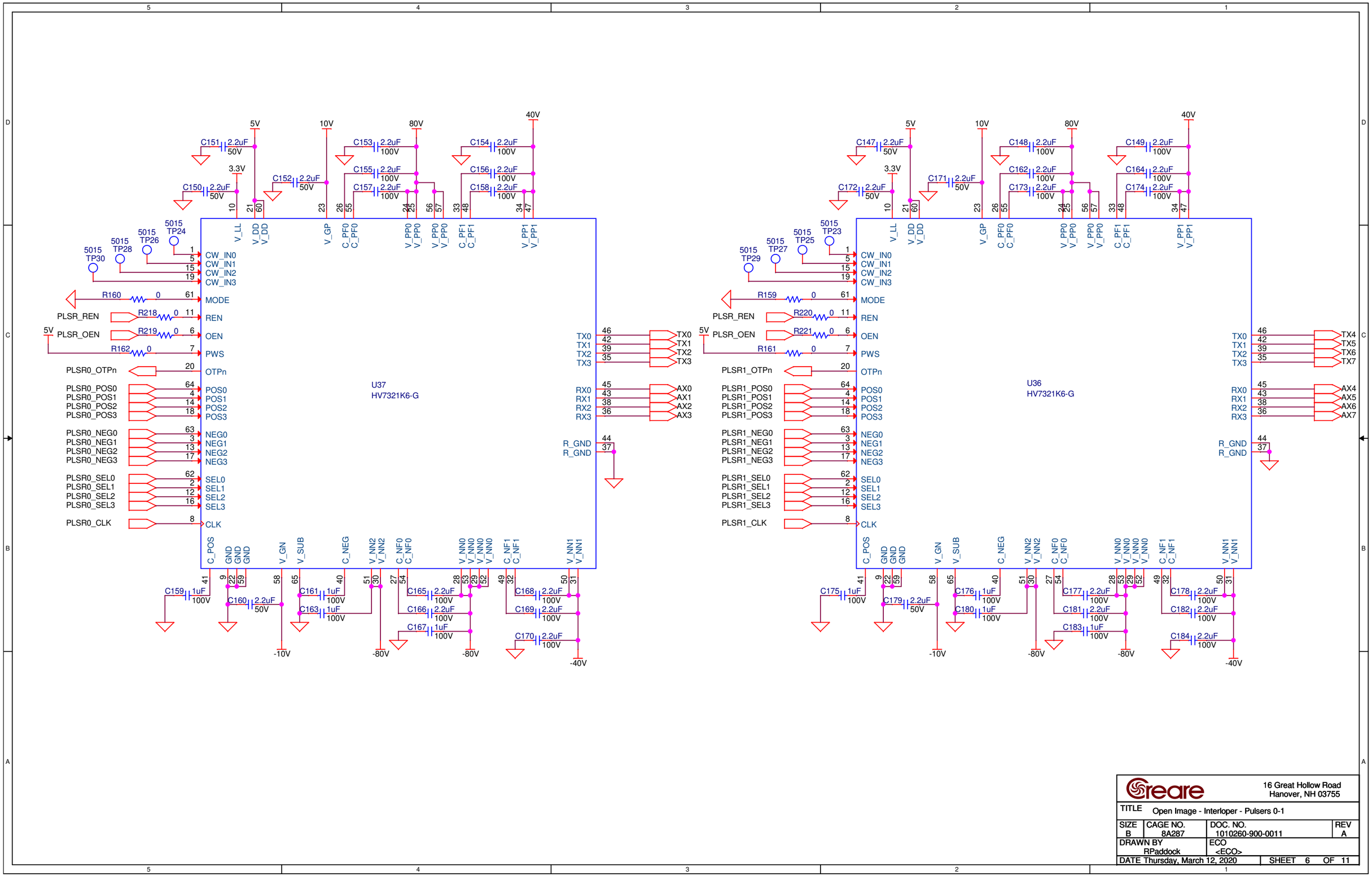




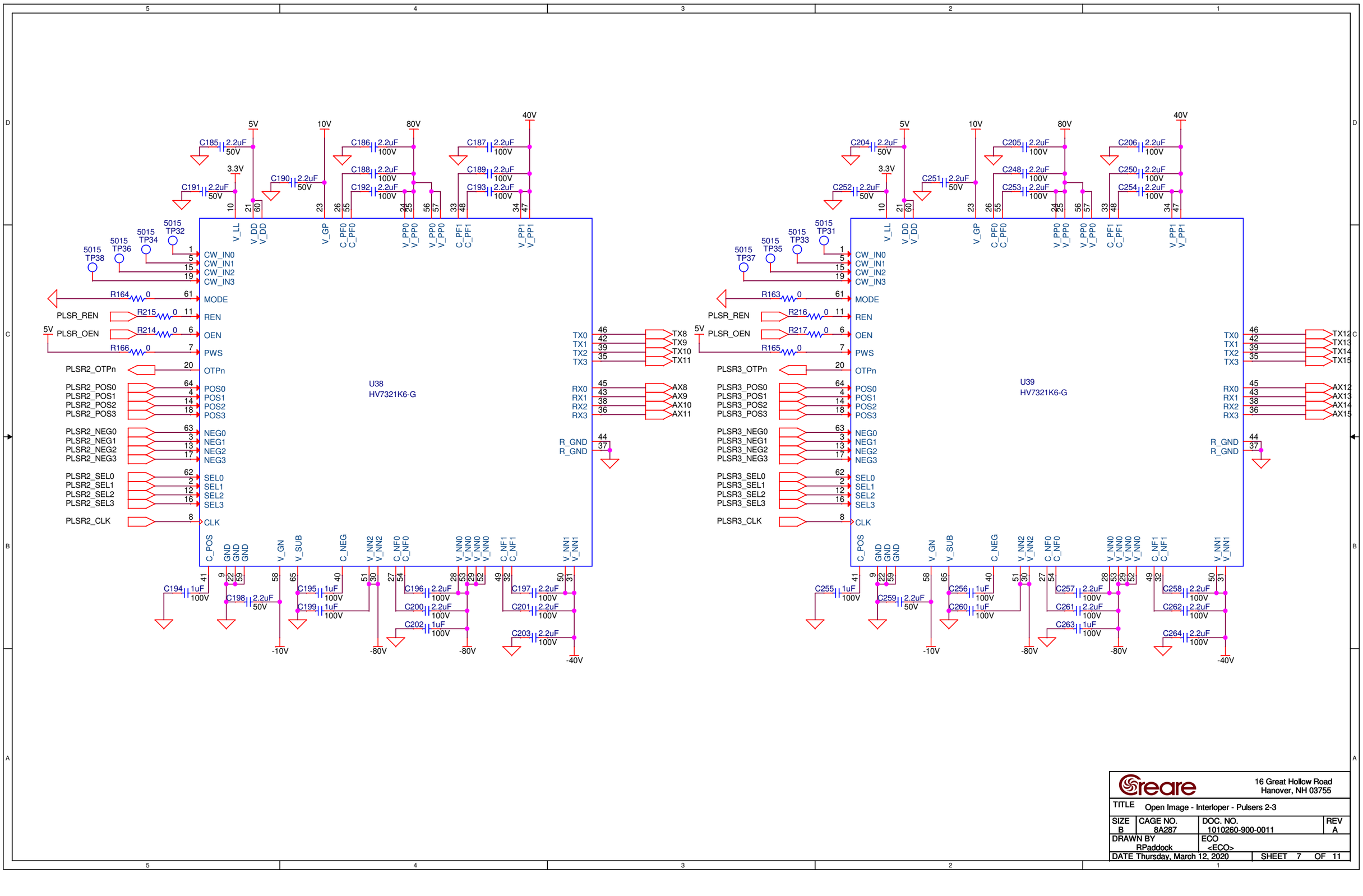


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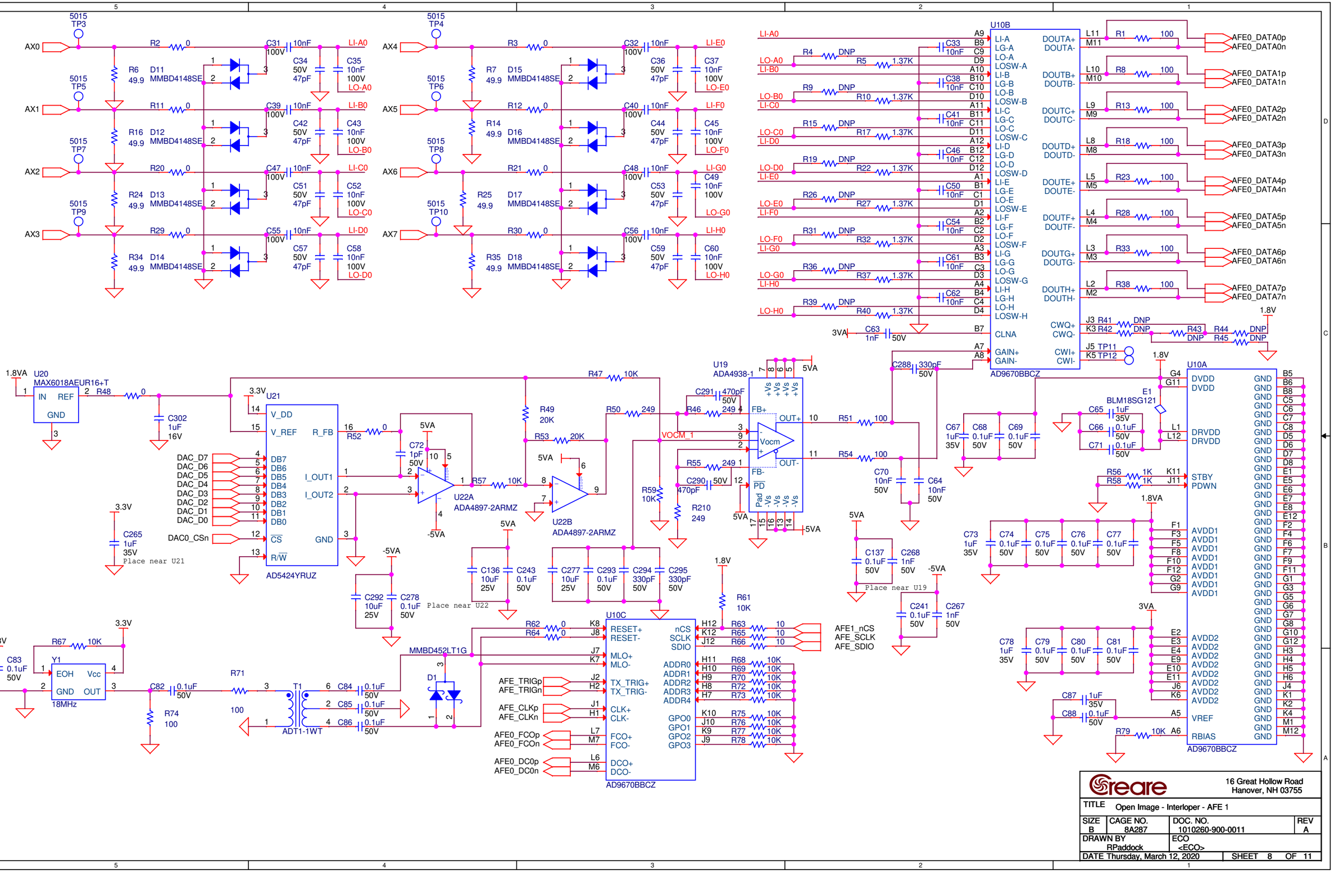
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SIZE B	CAGE NO. 8A287	DOC. NO. 1010260-900-0011	REV A
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DATE Tuesday, March 03, 2020		SHEET 5 OF 11	

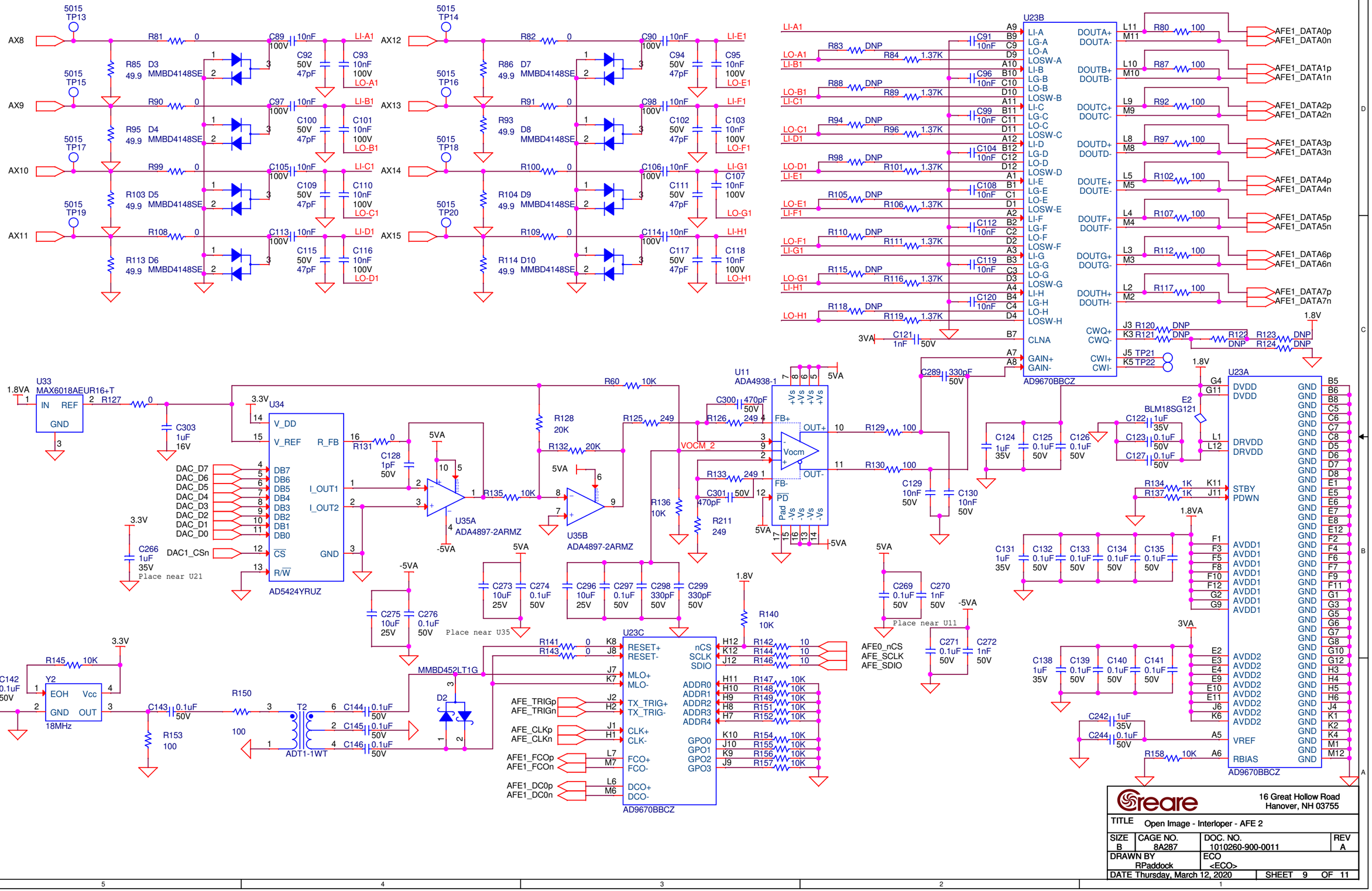


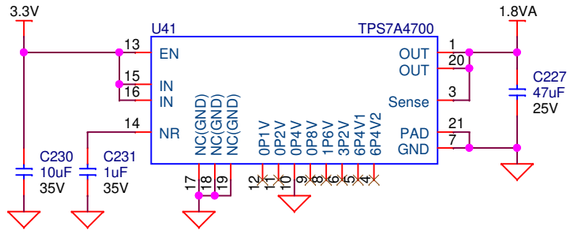
Creare	16 Great Hollow Road Hanover, NH 03755		
TITLE	Open Image - Interloper - Pulsers 0-1		
SIZE B	CAGE NO. 8A287	DOC. NO. 1010260-900-0011	REV A
DRAWN BY RPaddock		ECO <ECO>	
DATE Thursday, March 12, 2020		SHEET 6 OF 11	

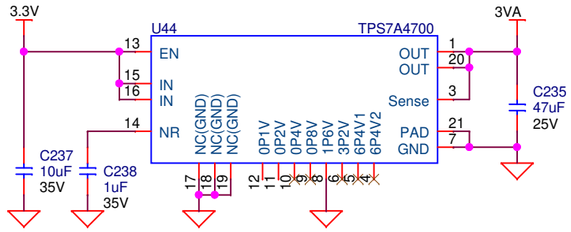


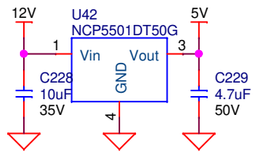
Creare	16 Great Hollow Road Hanover, NH 03755		
TITLE	Open Image - Interloper - Pulsers 2-3		
SIZE B	CAGE NO. 8A287	DOC. NO. 1010260-900-0011	REV A
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DATE Thursday, March 12, 2020		SHEET 7 OF 11	

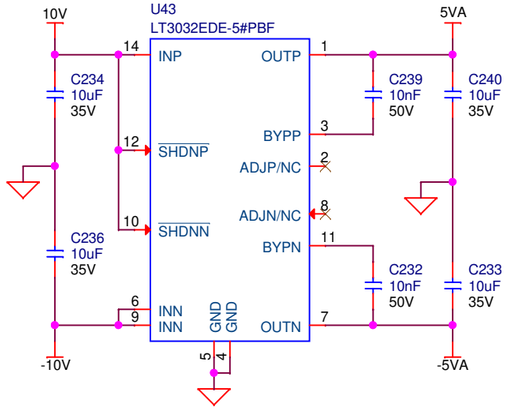

AX0
AX1
AX2
AX3
AX4
AX5
AX6
AX7
5015 TP3
5015 TP5
5015 TP7
5015 TP9
5015 TP4
5015 TP6
5015 TP8
5015 TP10
R2 0
R3 0
R11 0
R12 0
R20 0
R21 0
R29 0
R30 0
R6 49.9
R7 49.9
R16 49.9
R14 49.9
R24 49.9
R25 49.9
R34 49.9
R35 49.9
D11 MMBD4148SE
D15 MMBD4148SE
D12 MMBD4148SE
D16 MMBD4148SE
D13 MMBD4148SE
D17 MMBD4148SE
D14 MMBD4148SE
D18 MMBD4148SE
C31 10nF 100V
C32 10nF 100V
C39 10nF 100V
C40 10nF 100V
C47 10nF 100V
C48 10nF 100V
C55 10nF 100V
C56 10nF 100V
C34 50V 47pF
C36 50V 47pF
C42 50V 47pF
C44 50V 47pF
C51 50V 47pF
C53 50V 47pF
C57 50V 47pF
C59 50V 47pF
C35 10nF 100V
C37 10nF 100V
C43 10nF 100V
C45 10nF 100V
C52 10nF 100V
C49 10nF 100V
C58 10nF 100V
C60 10nF 100V
LI-A0
LI-E0
LI-B0
LI-F0
LI-C0
LI-G0
LI-D0
LI-H0
LO-A0
LO-E0
LO-B0
LO-F0
LO-C0
LO-G0
LO-D0
LO-H0
U10B
AD9670BBCZ
LI-A LG-A LO-A LOSW-A
LI-B LG-B LO-B LOSW-B
LI-C LG-C LO-C LOSW-C
LI-D LG-D LO-D LOSW-D
LI-E LG-E LO-E LOSW-E
LI-F LG-F LO-F LOSW-F
LI-G LG-G LO-G LOSW-G
LI-H LG-H LO-H LOSW-H
CLNA
GAIN+
GAIN-
R4 DNP
R5 1.37K
R9 DNP
R10 1.37K
R15 DNP
R17 1.37K
R19 DNP
R22 1.37K
R26 DNP
R27 1.37K
R31 DNP
R32 1.37K
R36 DNP
R37 1.37K
R39 DNP
R40 1.37K
C33 10nF
C38 10nF
C41 10nF
C46 10nF
C50 10nF
C54 10nF
C61 10nF
C62 10nF
C63 1nF 50V
3VA
DOUTA+ DOUTA-
DOUTB+ DOUTB-
DOUTC+ DOUTC-
DOUTD+ DOUTD-
DOUTE+ DOUTE-
DOUTF+ DOUTF-
DOUTG+ DOUTG-
DOUTH+ DOUTH-
CWQ+ CWQ-
CWI+ CWI-
L11 M11 R1 100
L10 M10 R8 100
L9 M9 R13 100
L8 M8 R18 100
L5 M5 R23 100
L4 M4 R28 100
L3 M3 R33 100
L2 M2 R38 100
AFE0_DATA0p AFE0_DATA0n
AFE0_DATA1p AFE0_DATA1n
AFE0_DATA2p AFE0_DATA2n
AFE0_DATA3p AFE0_DATA3n
AFE0_DATA4p AFE0_DATA4n
AFE0_DATA5p AFE0_DATA5n
AFE0_DATA6p AFE0_DATA6n
AFE0_DATA7p AFE0_DATA7n
J3 K3 R41 DNP R42 DNP R43 DNP R44 DNP R45 DNP
J5 K5 TP11 TP12
1.8V
1.8VA
U20
MAX6018AEUR16+T
IN REF GND
R48 0
C302 1uF 16V
3.3V
U21
AD5424YRUZ
V_DD V_REF R_FB I_OUT1 I_OUT2 GND
DB7 DB6 DB5 DB4 DB3 DB2 DB1 DB0
CS R/W
DAC_D7 DAC_D6 DAC_D5 DAC_D4 DAC_D3 DAC_D2 DAC_D1 DAC_D0
DAC0_CSn
R52 0
C72 1pF 50V
R47 10K
R49 20K
R53 20K
R50 249
R57 10K
U22A ADA4897-2ARMZ
U22B ADA4897-2ARMZ
5VA
-5VA
C265 1uF 35V Place near U21
C292 10uF 25V
C278 0.1uF 50V
C136 10uF 25V
C243 0.1uF 50V
C277 10uF 25V
C293 0.1uF 50V
C294 330pF 50V
C295 330pF 50V
Place near U22
U19
ADA4938-1
+Vs +Vs +Vs +Vs
Vocm
FB+ FB- OUT+ OUT- PD Pad -Vs -Vs -Vs -Vs
C291 470pF 50V
C290 470pF 50V
R46 249
R55 249
R59 10K
R210 249
VOCM_1
R51 100
R54 100
C288 330pF 50V
C70 10nF 50V
C64 10nF 50V
C67 1uF 35V
C68 0.1uF 50V
C69 0.1uF 50V
C137 0.1uF 50V
C268 1nF 50V
Place near U19
C241 0.1uF 50V
C267 1nF 50V
AFE1_nCS
AFE_SCLK
AFE_SDIO
U10C
AD9670BBCZ
RESET+ RESET- MLO+ MLO- TX_TRIG+ TX_TRIG- CLK+ CLK- FCO+ FCO- DCO+ DCO-
nCS SCLK SDIO ADDR0 ADDR1 ADDR2 ADDR3 ADDR4 GPO0 GPO1 GPO2 GPO3
R61 10K
R62 0
R64 0
R63 10
R65 10
R66 10
R68 10K
R69 10K
R70 10K
R72 10K
R73 10K
R75 10K
R76 10K
R77 10K
R78 10K
AFE_TRIGp AFE_TRIGn
AFE_CLKp AFE_CLKn
AFE0_FCOp AFE0_FCOn
AFE0_DC0p AFE0_DC0n
MMBD452LT1G
D1
U10A
AD9670BBCZ
DVDD DRVDD AVDD1 AVDD2 VREF RBIAS STBY PDWN
GND
E1 BLM18SG121
C65 1uF 35V
C66 0.1uF 50V
C71 0.1uF 50V
C73 1uF 35V
C74 0.1uF 50V
C75 0.1uF 50V
C76 0.1uF 50V
C77 0.1uF 50V
C78 1uF 35V
C79 0.1uF 50V
C80 0.1uF 50V
C81 0.1uF 50V
C87 1uF 35V
C88 0.1uF 50V
R56 1K
R58 1K
R79 10K
3VA
18MHz
Y1
EOH Vcc GND OUT
C83 0.1uF 50V
R67 10K
C82 0.1uF 50V
R74 100
R71 100
T1
ADT1-1WT
C84 0.1uF 50V
C85 0.1uF 50V
C86 0.1uF 50V
Creare
16 Great Hollow Road
Hanover, NH 03755
TITLE Open Image - Interloper - AFE 1
SIZE B
CAGE NO. 8A287
DOC. NO. 1010260-900-0011
REV A
DRAWN BY RPaddock
ECO <ECO>
DATE Thursday, March 12, 2020
SHEET 8 OF 11

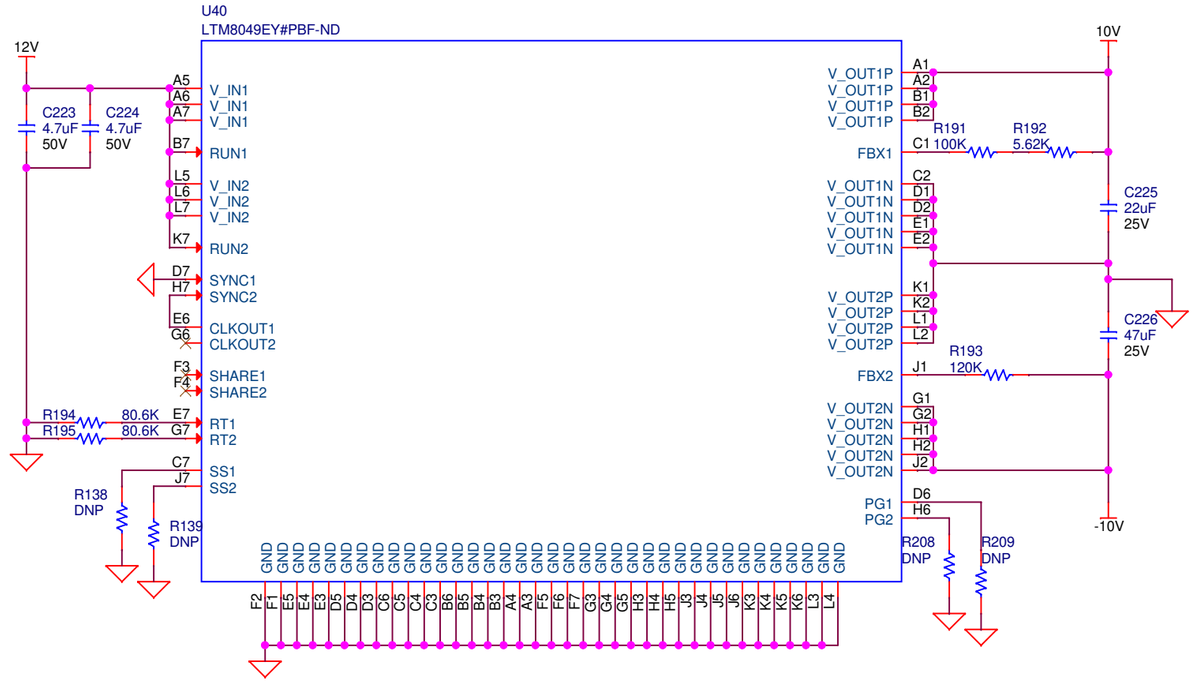

AX8
AX9
AX10
AX11
AX12
AX13
AX14
AX15
R81 0
R82 0
R90 0
R91 0
R99 0
R100 0
R108 0
R109 0
R85 49.9
R86 49.9
R93 49.9
R95 49.9
R104 49.9
R103 49.9
R113 49.9
R114 49.9
D3 MMBD4148SE
D7 MMBD4148SE
D4 MMBD4148SE
D8 MMBD4148SE
D5 MMBD4148SE
D9 MMBD4148SE
D6 MMBD4148SE
D10 MMBD4148SE
C89 10nF 100V
C90 10nF 100V
C97 10nF 100V
C98 10nF 100V
C105 10nF 100V
C106 10nF 100V
C113 10nF 100V
C114 10nF 100V
C92 50V 47pF
C94 50V 47pF
C100 50V 47pF
C102 50V 47pF
C109 50V 47pF
C111 50V 47pF
C115 50V 47pF
C117 50V 47pF
C93 10nF 100V LO-A1
C95 10nF 100V LO-E1
C101 10nF 100V LO-B1
C103 10nF 100V LO-F1
C110 10nF 100V LO-C1
C107 10nF 100V LO-G1
C116 10nF 100V LO-D1
C118 10nF 100V LO-H1
LI-A1
LI-E1
LI-B1
LI-F1
LI-C1
LI-G1
LI-D1
LI-H1
5015 TP13
5015 TP14
5015 TP15
5015 TP16
5015 TP17
5015 TP18
5015 TP19
5015 TP20
U23B
AD9670BBCZ
LI-A
LG-A
LO-A
LOSW-A
LI-B
LG-B
LO-B
LOSW-B
LI-C
LG-C
LO-C
LOSW-C
LI-D
LG-D
LO-D
LOSW-D
LI-E
LG-E
LO-E
LOSW-E
LI-F
LG-F
LO-F
LOSW-F
LI-G
LG-G
LO-G
LOSW-G
LI-H
LG-H
LO-H
LOSW-H
CLNA
GAIN+
GAIN-
DOUTA+
DOUTA-
DOUTB+
DOUTB-
DOUTC+
DOUTC-
DOUTD+
DOUTD-
DOUTE+
DOUTE-
DOUTF+
DOUTF-
DOUTG+
DOUTG-
DOUTH+
DOUTH-
CWQ+
CWQ-
CWI+
CWI-
AFE1_DATA0p
AFE1_DATA0n
AFE1_DATA1p
AFE1_DATA1n
AFE1_DATA2p
AFE1_DATA2n
AFE1_DATA3p
AFE1_DATA3n
AFE1_DATA4p
AFE1_DATA4n
AFE1_DATA5p
AFE1_DATA5n
AFE1_DATA6p
AFE1_DATA6n
AFE1_DATA7p
AFE1_DATA7n
R80 100
R87 100
R92 100
R97 100
R102 100
R107 100
R112 100
R117 100
R83 DNP
R84 1.37K
R88 DNP
R89 1.37K
R94 DNP
R96 1.37K
R98 DNP
R101 1.37K
R105 DNP
R106 1.37K
R110 DNP
R111 1.37K
R115 DNP
R116 1.37K
R118 DNP
R119 1.37K
C91 10nF
C96 10nF
C99 10nF
C104 10nF
C108 10nF
C112 10nF
C119 10nF
C120 10nF
C121 1nF 50V
3VA
1.8V
J3 R120 DNP
K3 R121 DNP
R122 DNP
R123 DNP
R124 DNP
J5 TP21
K5 TP22
U33
MAX6018AEUR16+T
1.8VA
IN REF GND
R127 0
C303 1uF 16V
3.3V
U34
V_DD V_REF R_FB I_OUT1 I_OUT2 GND
DB7 DB6 DB5 DB4 DB3 DB2 DB1 DB0
CS R/W
AD5424YRUZ
DAC_D7 DAC_D6 DAC_D5 DAC_D4 DAC_D3 DAC_D2 DAC_D1 DAC_D0
DAC1_CSn
C266 1uF 35V Place near U21
R131 0
C128 1pF 50V
U35A ADA4897-2ARMZ
U35B ADA4897-2ARMZ
5VA
-5VA
R135 10K
R128 20K
R132 20K
R125 249
R126 249
R136 10K
R211 249
U11 ADA4938-1
C300 470pF 50V
C301 470pF 50V
R133 249
R129 100
R130 100
C129 10nF 50V
C130 10nF 50V
C289 330pF 50V
C273 10uF 25V
C274 0.1uF 50V
C296 10uF 25V
C297 0.1uF 50V
C298 330pF 50V
C299 330pF 50V
C275 10uF 25V
C276 0.1uF 50V
Place near U35
C269 0.1uF 50V
C270 1nF 50V
Place near U11
C271 0.1uF 50V
C272 1nF 50V
AFE0_nCS
AFE_SCLK
AFE_SDIO
R140 10K
U23C
AD9670BBCZ
RESET+ RESET- MLO+ MLO- TX_TRIG+ TX_TRIG- CLK+ CLK- FCO+ FCO- DCO+ DCO-
nCS SCLK SDIO ADDR0 ADDR1 ADDR2 ADDR3 ADDR4 GPO0 GPO1 GPO2 GPO3
R141 0 R143 0
R142 10 R144 10 R146 10
R147 10K R148 10K R149 10K R151 10K R152 10K
R154 10K R155 10K R156 10K R157 10K
AFE_TRIGp AFE_TRIGn
AFE_CLKp AFE_CLKn
AFE1_FCOp AFE1_FCOn
AFE1_DC0p AFE1_DC0n
MMBD452LT1G
D2
C142 0.1uF 50V
R145 10K
Y2
EOH Vcc GND OUT
18MHz
C143 0.1uF 50V
R153 100
R150 100
T2
ADT1-1WT
C144 0.1uF 50V
C145 0.1uF 50V
C146 0.1uF 50V
U23A
AD9670BBCZ
DVDD DRVDD STBY PDWN AVDD1 AVDD2 VREF RBIAS GND
E2 BLM18SG121
C122 1uF 35V
C123 0.1uF 50V
C127 0.1uF 50V
C124 1uF 35V
C125 0.1uF 50V
C126 0.1uF 50V
R134 1K
R137 1K
C131 1uF 35V
C132 0.1uF 50V
C133 0.1uF 50V
C134 0.1uF 50V
C135 0.1uF 50V
C138 1uF 35V
C139 0.1uF 50V
C140 0.1uF 50V
C141 0.1uF 50V
C242 1uF 35V
C244 0.1uF 50V
R158 10K
G4 G11
L11 M11
L10 M10
L9 M9
L8 M8
L5 M5
L4 M4
L3 M3
L2 M2
L1 L12
K11 J11
F1 F3 F5 F8 F10 F12 G2 G9
E2 E3 E4 E9 E10 E11 J6 K6
A5
A6
B5 B6 B8 C5 C6 C7 C8 D5 D6 D7 D8 E1 E5 E6 E7 E8 E12 F2 F4 F6 F7 F9 F11 G1 G3 G5 G6 G7 G8 G10 G12 H3 H4 H5 H6 J4 K1 K2 K4 M1 M12
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16 Great Hollow Road
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TITLE Open Image - Interloper - AFE 2
SIZE B
CAGE NO. 8A287
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DATE Thursday, March 12, 2020
SHEET 9 OF 11

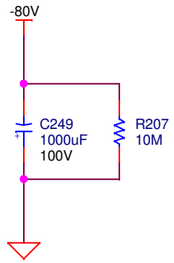


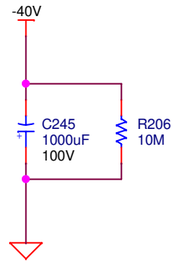


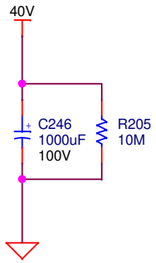


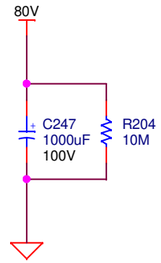


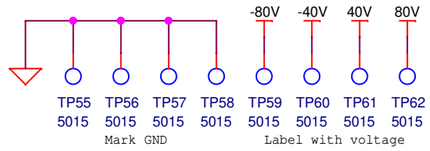




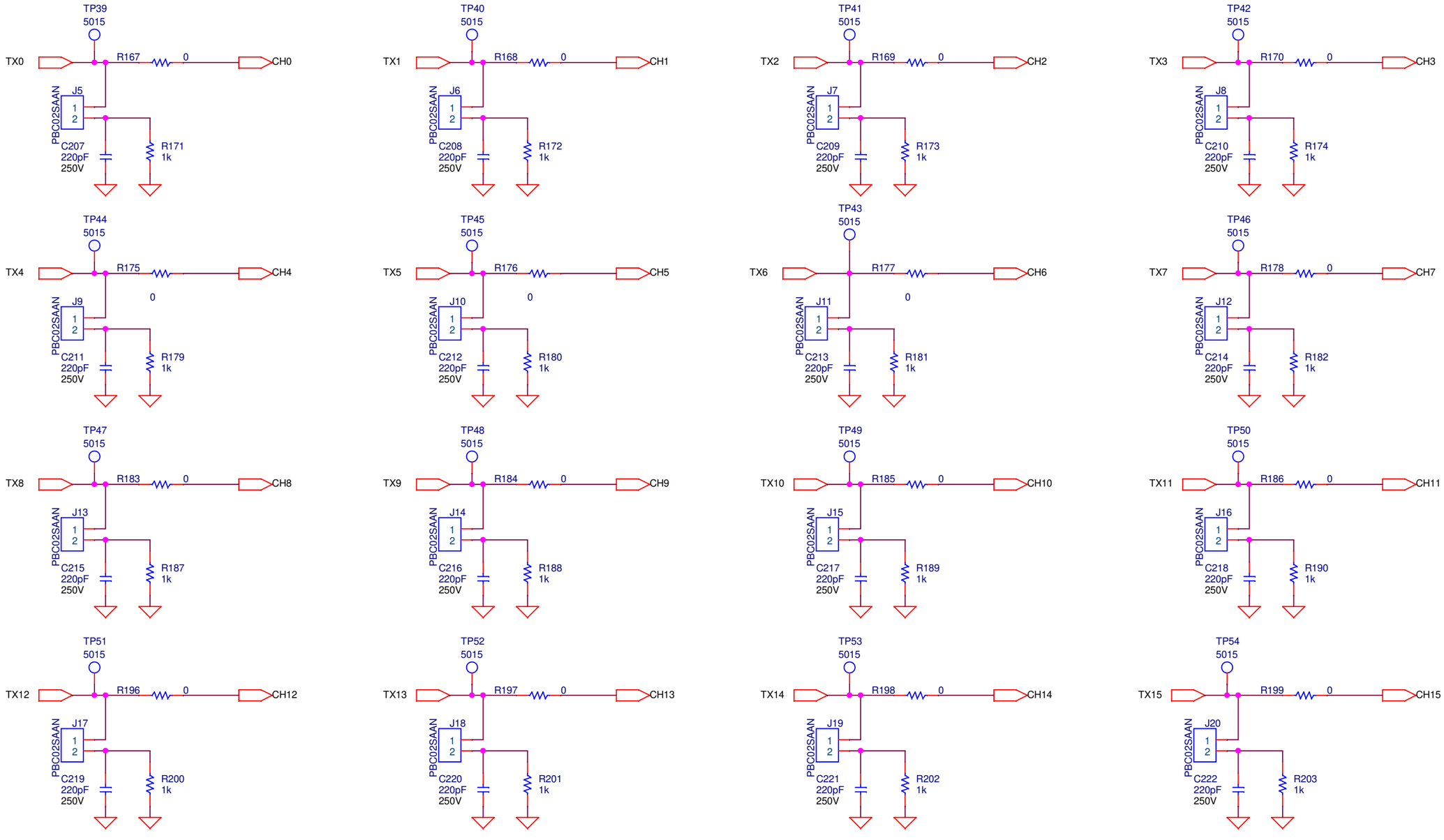








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